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Translational Population-Pharmacodynamic Modeling of a Novel Long-Acting siRNA Therapy, Inclisiran, for the Treatment of Hypercholesterolemia

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Conflict of Interest statement:

Nathalie H Gosselin, Virna J.A. Schuck, Olivier Barriere, and Patrick Smith are employees of Certara, a consulting firm in integrated drug development, and have directly consulted with pharmaceutical companies in the development of inclisiran. YanLing He, Kenneth Kulmatycki, Alison Margolskee are employees of Novartis Pharmaceuticals.

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Abstract

Inclisiran is a novel N-acetylgalactosamine (GalNAc) conjugated small-interfering ribonucleic acid (siRNA) therapy designed to specifically target proprotein convertase subtilisin/kexin type 9 (PCSK9) mRNA in the liver for the treatment of hypercholesterolemia. Inclisiran's GalNAc attachment results in a rapid uptake into the liver, and thus a short plasma half-life, but long duration of effects on PCSK9 inhibition and low-density lipoprotein cholesterol (LDL-C) lowering. The effects on PCSK9 inhibition and consequent LDL-C reduction are sustained for more than six months following a single subcutaneous (SC) dose, despite inclisiran being detectable in the plasma only for up to 48 hours. A kinetic-pharmacodynamic (K-PD) model was developed to characterize inclisiran's dose-related LDL-C lowering effects and to evaluate the impact of intrinsic and extrinsic factors on LDL-C lowering. To accommodate the long duration of action, the K-PD model incorporated an effect compartment which represents the liver. Inclisiran concentration in the liver leads to decreased production of the PCSK9 protein and allow recycling of more LDL-C receptors on the hepatocyte cell surface, which results in a reduction of circulating LDL-C. The analysis of covariates identified PCSK9 and LDL-C baseline levels as important factors for the effects of LDL-C lowering. Observations and modeling and simulation results demonstrated that PCSK9 and LDL-C reductions are achieved rapidly after dosing and sustained when patients are treated with a 300 mg SC dose once every 6 months.

Introduction

Atherosclerotic cardiovascular disease (ASCVD) is the leading cause of death and disability worldwide and is expected to remain so beyond 2040.¹ After over 30 years of developing innovative pharmacotherapy for the treatment of this disease, death rates associated with ASCVD are still increasing in the United States and worldwide.² This may be due to the multiple factors that contribute to the pathophysiology and development of ASCVD. However, strong and consistent evidence has established that reducing low-density lipoprotein cholesterol (LDL-C) below the treatment targets (LDL-C<70 mg/dL in USA³; LDL-C<55 mg/dL in EU⁴) is critical in decreasing the cardiovascular risk for patients with ASCVD, and for the prevention of recurring cardiovascular events. LDL-C is not merely a biomarker of increased risk, but a causal and modifiable factor in the pathogenesis of ASCVD.⁵

As mentioned, despite significant progress over the last 30 years in developing LDL-C lowering therapies, more than 50% of the patients with ASCVD do not reach the treatment target of LDL-C <70 mg/dL,⁶ therefore, significant unmet need remains for pharmacotherapy that further reduces LDL-C, particularly in combination with statins, which is the first line therapy for both primary and secondary prevention of ASCVD. The lack of sufficient LDL-C lowering effects with current therapies includes factors such as muscular adverse effects associated with statins, low adherence with daily oral administration for a long period of lifetime, and lack of response to currently available drugs. Simulation models of access, prescribing, dispensing, adherence, and persistence based on population data show that a substantial number of patients need more effective treatment options and long-term adherence is a major factor contributing to the lack of response or effectiveness.⁷

Abifadel et al demonstrated the critical role of proprotein convertase subtilisin/kexin type 9 (PCSK9) in regulating hepatic low-density lipoprotein receptor (LDL-R) activity in 2003.⁸ The PCSK9 protein binds to LDLRs, targeting them for lysosomal degradation; leading to reduced LDLR on the hepatocytes cell membrane and increased serum LDL-C concentrations.⁹ PCSK9 as a target for reducing LDL-C has been validated both by genetic evidence in humans and pharmacotherapy with anti-PCSK9 monoclonal antibodies.^{8,10}

Inclisiran is a novel small-interfering ribonucleic acid (siRNA) therapy that is designed to specifically target PCSK9 mRNA in the liver for the treatment of hypercholesterolemia.

Inclisiran is a chemically modified double stranded small interfering ribonucleic acid (RNA) conjugated on the sense strand with triantennary N-acetylgalactosamine (GalNAc) to facilitate uptake by hepatocytes.¹¹ Following a subcutaneous (SC) injection of inclisiran, the GalNAc-conjugated siRNA is rapidly taken up into the hepatocytes, the antisense strand is then incorporated in the RNA-induced silencing complex (RISC) and directs the catalytic breakdown of mRNA for PCSK9. This inhibits the translation of PCSK9 mRNA to the PCSK9 protein. The reduced PCSK9 protein levels lead to increased LDL-C receptor recycling and expression on the hepatocyte cell surface, resulting in increased LDL-C uptake to the liver and reduced LDL-C levels in the systemic circulation. Inclisiran is primarily metabolized by exonuclease and endonuclease and excreted in the urine. While inclisiran is detectable in plasma for only 24 to 48 hours after SC administration, the effects on PCSK9 inhibition and plasma LDL-C lowering last for months after a single dose administration.¹² With this characteristic of GalNAc-conjugated siRNA, systemic exposure of inclisiran is disconnected to their PD effects on PCSK9 inhibition and the LDL-C lowering effects. Inclisiran has been approved in US, EU and more than 50 other countries worldwide for the treatment of hypercholesterolemia in patients with heterozygous

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familial hypercholesterolemia (HeFH), ASCVD or ASCVD-risk equivalent on top of the maximal tolerated statin therapy, or as monotherapy for patients who are intolerant of statins.¹³

During the clinical development of inclisiran, multiple dose-ranging and clinical pharmacology studies were conducted in healthy subjects, and pivotal phase II and III clinical trials in patients. The initial clinical data obtained in the Phase II study were leveraged to develop a preliminary population kinetic-pharmacodynamic (K-PD) model to select the inclisiran dose and dose regimen to be tested in Phase III clinical trials. A K-PD model (rather than a pharmacokinetic (PK)/PD model) was established due to the temporal disconnect between inclisiran systemic exposure (detectable in blood up to 48 hours) and PD response (sustained for >6 months). The model was subsequently updated as additional clinical data from Phase III trials became available to support inclisiran dose optimization and subsequent approval of inclisiran by both EMA in 2020 and FDA in 2021. The primary objective of this manuscript is to describe the modeling approach and demonstrate the utility of the population K-PD model to support the determination of dosing regimen and clinical dose recommendations for inclisiran. The population K-PD model for inclisiran was developed based on data from more than 4000 subjects with ASCVD or HeFH and healthy volunteers across multiple Phase I to III clinical trials.

Methods

Clinical Trials

Multiple clinical trials were conducted during the development of inclisiran, which collected PD data (ie, PCSK9 and LDL-C) following SC administration of 25 mg to 900 mg doses of inclisiran. A total of 9 studies were included in the analysis, four Phase 1 studies (ALNPCSSC-001, ORION-6, ORION-7, and ORION-12), two Phase 2 (ORION-1 and ORION-3 [up to Day

630]), and three Phase 3 (ORION-9, ORION-10, and ORION-11). These studies enrolled different populations, including healthy subjects, subjects with renal or hepatic impairment, subjects with elevated LDL-C, with HeFH (from ORION-9 study), and with ASCVD or ASCVD-risk equivalents (from ORION-1, ORION-3, ORION-10 and ORION-11) (referred to in this manuscript as ASCVD population).

A total of 221 patients with ASCVD or ASCVD-risk equivalents in ORION-10 and ORION-11 was classified as familial hypercholesterolemia (FH) (7% of 3174 patients in ORION-10 and ORION-11) and amongst those FH patients, 48 were flagged as heterozygous familial hypercholesterolemia (HeFH) (1.5% of 3174 patients). The FH status was unknown for 1165 patients (37%) in ORION-10 and ORION-11. In the current analysis, only patients from ORION-9 were included in the HeFH population and in ORION-10 and ORION-11, the 2971 patients with ASCVD (93.6%) were regrouped with the 203 patients with ASCVD-risk equivalents (6.4%).

All study protocols, amendments, and consent forms were approved by the relevant Institutional Review Board or Independent Ethics Committee. The studies were conducted in accordance with the ethical principles that have their origins in the Declaration of Helsinki and in compliance with Good Clinical Practice (GCP). All participants provided written informed consent.

PCSK9 and LDL-C Data

Blood samples for assessment of PCSK9 and LDL-C before and during treatment with inclisiran were collected over the duration of treatment. PD data collected in the clinical studies were combined for the development of the population K-PD model. A rich PD data set was collected in all studies. The large number of samples (30,476 and 46,634 samples with PCSK9 and LDL-C, respectively) at several timepoints for a long duration (up to 3 years) resulted to slow

computational processes to achieve model convergence. Therefore, the model development was facilitated by excluding data according to a more restrictive exclusion criteria (exclusion criteria #1) and then the final model was re-run based on data with less restrictive criteria (exclusion criteria #2), as listed below.

Exclusion criteria #1: Subjects with PCSK9 at baseline >1200 ng/mL and/or LDL-C at baseline >400 mg/dL; PCSK9 samples >1200 ng/mL during trials; LDL-C samples >400 mg/dL during trial; subjects with treatment deviation with inconsistency in PD profiles such as subjects who received placebo with significant decreased of PCSK9 and LDL-C (i.e., 146 excluded samples with PCSK9 levels over 33,349 [0.44%] and 151 excluded samples with LDL-C levels over 43,761 [0.35%]).

Exclusion criteria #2: Subjects with PCSK9 at baseline >2000 ng/mL; PCSK9 samples >2000 ng/mL at any time; LDL-C samples >600 mg/dL at any time; Subjects with treatment deviation with inconsistency in PD profiles (i.e., 39 excluded samples with PCSK9 levels over 33,349 [0.12%] and 27 excluded samples with LDL-C levels over 43,761 [0.062%]).

Data Description

The PD analysis dataset included 77,110 measurable PD data points from 4328 subjects for PCSK9 and LDL-C. The PD population included 1967 subjects in the placebo group (12,832 and 17,644 samples for PCSK9 and LDL-C, respectively) and 2361 subjects in the inclisiran group (20,517 and 26,117 samples for PCSK9 and LDL-C, respectively).

Among the 77,110 PD observations, a total of 297 measurable PD observations (ie, 0.4%) were excluded from the analysis based on exclusion criteria #1, and 66 measurable PD observations (ie, 0.1%) were excluded based on exclusion criteria #2. The proportions of excluded samples in

both treatments are negligible for both analytes (<0.5%). Of the 4328 subjects included in the PD analysis dataset, 14 subjects were excluded from the analysis based on exclusion criteria #1 and 5 subjects based on exclusion criteria #2.

Population K-PD model

The base structure of the models to describe the effect of inclisiran on PCSK9 and LDL-C was a turnover model.¹⁴ The hypothetical liver effect compartment of inclisiran was linked to the zero-order synthesis rate for PCSK9 through an Imax model. PCSK9 levels were then linked to the first order degradation/clearance rate of LDL-C using a second Imax model. A schematic representation of the base structural model is presented in Figure 1.

The model is described with the following differential equations:

$$\frac{dCe}{dt} = -Ce \times Ke$$

$$\frac{d[PCSK9]}{dt} = Ksyn_p \times \left(1 - \frac{Imax_p \times Ce^{\gamma_p}}{Ce^{\gamma_p} + IC50_p^{\gamma_p}}\right) - Kdeg_p \times [PCSK9]$$

$$\frac{d[LDL-C]}{dt} = Ksyn_L - \left(1 - \frac{Imax_L \times [PCSK9]^{\gamma_L}}{[PCSK9]^{\gamma_L} + IC50_L^{\gamma_L}}\right) \times Kdeg_L \times [LDL-C]$$

where Ce is the amount of inclisiran in a hypothetical liver effect compartment; Ke is the first-order rate constant for the effect compartment; Imax_p and Imax_L are the maximum inhibitory effects of inclisiran on PCSK9 and LDL-C levels, respectively; γ_p and γ_L are Hill coefficients on PCSK9 and LDL-C, respectively; Kdeg_p is the degradation rate of PCSK9; Kdeg_L is the degradation rate of LDL-C; Ksyn_p is the synthesis rate of PCSK9; and Ksyn_L is the synthesis rate of LDL-C. The concentrations of PCSK9 and LDL-C are described by [PCSK9] and [LDL-C], respectively.

The population K-PD analyses and Monte-Carlo simulations were performed using nonlinear mixed effect modeling analysis in nonlinear mixed-effects modeling software (NONMEM®) (Version 7.4). Dataset exploration and figures were performed using R® (Version 3.5).

Results

Population characteristics

Baseline values of continuous and categorical covariates stratified by population are summarized in Table 1.

The means for body weight in subjects with ASCVD and HeFH were slightly higher than the mean value in the healthy subjects (15% and 9% higher in subjects with ASCVD and HeFH, respectively compared to healthy subjects). Subjects with ASCVD tended to be older, with mean age of 65.1 years, compared to 54.7 years for subjects with HeFH from ORION-9.

The PD population was 67.2% male, primarily of White origin (91.4%). The proportion of male subjects in the HeFH population was slightly lower (47.2%). The majority of subjects with ASCVD and HeFH were nondiabetic (ie, 72.6% and 91.1%, respectively) and were taking a statin at baseline (81.5% and 74.8%, respectively).

Baseline levels of the PD measurements were similar in healthy subjects and in subjects with renal or hepatic impairment. Mean baseline PCSK9 values in subjects with ASCVD and HeFH were higher than the mean values in the Phase 1 population (37% and 54% higher in ASCVD and HeFH, respectively), whereas the mean baseline value of LDL-C in subjects with ASCVD and with HeFH was 15% lower and 23% higher, respectively, than the mean value in Phase 1 population. These differences are mostly likely the result of disease background and lipid-

lowering therapy. Subjects with ASCVD had higher use of lipid lowering therapy compared to the other populations.

Triglyceride levels were slightly higher in subjects with ASCVD and HeFH with mean values of 151 and 135 mg/dL, respectively, in comparison to mean values of between 111 and 128 mg/dL for the other populations.

Population K-PD model

The inclusion of the covariates on the PD parameters explained part of their inter-individual variability (IIV). Based on the dataset with exclusion criteria #1, the populations and body weight explained 4.5% of IIV on K_e (ie, IIV of 43.1% in the reduced covariate K-PD model [exclusion criteria #1] and 47.6% in the structural population K-PD model; Table S1). The inclusion of nine covariates on IC_{50L} decreased IIV by 11.8% (ie, 143.8% in the reduced covariate K-PD model [exclusion criteria #1] versus 155.6% in the structural population K-PD model; Table S1). Negligible impact was observed on IIV of IC_{50P} after including covariates with IIV of 89.1%.

Table 2 presents the PD estimates and covariate effects of the final population K-PD model with their uncertainty based on the dataset with exclusion criteria #2.

Goodness of Fit

The bootstrap of the final population K-PD model was run using exclusion criterion #2.

Parameter estimates and bootstrap CIs are reported in Table S2. Medians of PD parameters and covariate effects derived with the 502 bootstrap runs were close to the final parameters estimated with the original dataset (i.e., bias <20%), indicating the robustness of the estimations. All

covariate effects were statistically significant as the 95% CI did not include the null hypothesis (i.e., 0 for continuous and categorical covariates).

The basic goodness-of-fit plots of the final population K-PD model of PCSK9 and LDL-C after SC injection of inclisiran are presented in Figure S1. Figure S2 presents the prediction corrected VPC of the final population K-PD model for subjects in inclisiran and placebo groups. Observed medians and 2.5th and 97.5th percentiles for PCSK9 and LDL-C were generally within the 95% predictive interval (PI) of the simulated medians and simulated 2.5th and 97.5th percentiles, although a small bias was observed in the placebo group for subjects with higher levels of prediction-corrected PCSK9 and LDL C (ie, 97.5th percentile) at the end of the studies (Figure 2).

Covariate Effects

The covariate estimates derived with bootstrap runs of the final population K-PD model were used to derive the effect sizes (mean and 95% CI) of the change in PD parameters relative to the reference subject (i.e., a male 65 years of age, with a body weight of 87 kg and observed baseline PCSK9 and LDL-C values of 378.1 ng/mL and 100 mg/dL, respectively, nondiabetic, with ASCVD, with presence of statin, and with triglycerides of 130 mg/dL and bilirubin of 0.6 mg/dL). The impact of each relevant continuous covariate was evaluated based on the 2.5th and 97.5th percentiles of the covariate levels observed in Phase III studies. Categorical covariates were evaluated against the reference subject.

Magnitudes and uncertainties of covariate effects in annual means of LDL-C (raw levels, absolute changes, and percent changes) within each target population are presented in Figure 3, for subjects with ASCVD and for subjects with HeFH. Impacts of continuous covariates were evaluated based on the 2.5th and 97.5th percentiles observed within each population in Phase III

studies for subjects with ASCVD and for subjects with HeFH, considering exclusion criterion #2. Categorical covariates were evaluated against the subject with median values obtained within each population.

Magnitude of effect of covariates on annual mean levels of LDL-C during inclisiran treatment was considered clinically significant if the effect of the 2.5th or 97.5th percentiles of the covariates fell outside the 0.8 to 1.25 range compared to reference. Based on those criteria, only baseline values of PCSK9 and LDL-C may have clinically significant impact on annual mean levels of LDL-C during inclisiran treatment in subjects with ASCVD. More details of covariate impacts in subjects with ASCVD are described below:

- Annual mean levels of LDL-C increased with increasing baseline LDL-C: subjects with baseline LDL-C values of 60 mg/dL and 203 mg/dL are expected to have annual mean levels 1.26-fold lower and 1.60-fold higher relative to the reference subject with baseline LDL-C value of 96 mg/dL (i.e., 32.4 mg/dL and 69.8 mg/dL versus to 43.5 mg/dL, respectively).
- Annual mean levels of LDL-C decreased with increasing baseline PCSK9: subjects with baseline PCSK9 values of 200.6 ng/mL and 633.6 ng/mL are expected to have annual mean levels 1.34-fold higher and 1.23-fold lower relative to the reference subject, with baseline PCSK9 value of 371.8 ng/mL (i.e., 58.3 mg/dL and 33.4 mg/dL versus 43.5 ng/mL, respectively).
- Annual mean levels of LDL-C decreased with increasing triglyceride levels: subjects with triglycerides of 59 mg/dL and 348 mg/dL are expected to have annual mean levels 1.16-fold higher and 1.14-fold lower relative to the reference subject, with triglycerides of 132 mg/dL (i.e., 50.5 mg/dL and 37.2 mg/dL versus 43.5 mg/dL, respectively).

- Annual mean levels of LDL-C decreased with increasing age: subjects 47 and 81 years of age are expected to have LDL-C levels 1.15-fold higher and 1.09-fold lower in comparison to the reference subject, who was 66 years old (ie, 50.1 mg/dL and 40.0 mg/dL versus 43.5 mg/dL, respectively).
- Subjects without statins are expected to have 1.09-fold higher annual mean levels of LDL-C compared to subjects receiving statins (i.e., 47.6 mg/dL versus 43.5 mg/dL, respectively).

Simulations

Simulations of rich profiles of PCSK9 and LDL-C levels based on the final population K-PD model with IIV terms were performed in a virtual population (n=1000) receiving a 300 mg SC injection of inclisiran sodium on Day 1, on Day 90, and every 6 months (Figure 4). The virtual population was randomly selected among subject characteristics in Phase 3 studies. Due to the ~30% difference between baseline LDL-C levels across studies (mean of 153 mg/dL in ORION-9 and means of 105 mg/dL in both ORION-10 and ORION-11) and the different populations (ie, subjects with HeFH in ORION-9, and subjects with ASCVD in ORION-10 and ORION-11), the simulations were stratified by populations: ASCVD versus HeFH. Distributions of subject characteristics were obtained in a total of 500 virtual HeFH subjects resampled from ORION-9, 250 virtual subjects resampled from ORION-10, and another 250 subjects resampled from ORION-11. Medians and the 90% PI of the simulations were in good agreement with the observations.

Predicted steady-state levels of PCSK9 and LDL-C at pre-dose, nadirs, annual mean, and fluctuations, defined as trough-nadir, were derived based on individual predicted concentration-time profiles after the dose at Day 450 (Table 3). Steady-state simulations predicted lower

fluctuations of PCSK9 and LDL-C in subjects with ASCVD, in comparison to subjects with HeFH. The medians predicted in subjects with ASCVD were 56.3 ng/mL and 9.98 mg/dL for PCSK9 and LDL-C, respectively, versus 85.4 ng/mL and 16.6 mg/dL in HeFH, respectively. Similarly, AUEC levels differed by population, with medians of 198 ng×day/mL and 34.6 mg×day/dL for PCSK9 and LDL-C, respectively, in subjects with ASCVD, in comparison to 290 ng×day/mL and 55.2 mg×day/dL, respectively, in subjects with HeFH.

Simulations were also performed to evaluate the impact of missing or delayed doses on PCSK9 and LDL-C. Treatment response and percentage and absolute changes from baseline of PCSK9 and LDL-C were predicted for scenarios where a dose in the every-6-month regimen was either missed or given 1, 2, or 3 months later than scheduled. Figure 4 shows that delaying a 6-monthly scheduled dose by 1 to 3 months had a small impact on LDL-C levels, since subjects would rapidly return to optimal therapeutic levels once the delayed dose was administered. Due to the rapid time to reach steady state conditions, the delayed doses, once administered, resulted in very similar PCSK9 and LDL-C levels immediately prior to and following the next original scheduled dose. The simulations therefore demonstrate that if a 6-monthly dose is missed by less than 3 months, administer inclisiran and maintain dosing according to the patient's original schedule and if a planned dose is missed by more than 3 months, start a new dosing schedule - administer inclisiran initially, 3 months later, and then every 6 months.

Discussion

Inclisiran is a chemically synthesized, double stranded, small interfering ribonucleic acid (siRNA) oligonucleotide, with the sense strand covalently linked to a ligand comprising of 3 N-acetylgalactosamine (GalNAc) residues that enables specific uptake by hepatocytes.¹⁵ Inclisiran is only detectable in plasma for 24 to 48 hours after SC administration, however, it has

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demonstrated extended effects on PCSK9 inhibition and reduction of plasma LDL-C, which is most likely due to a very long half-life of inclisiran in the liver. The estimated half-life from the hypothetical effect compartment (liver) is in the range of approximately 80-90 days for subjects with ASCVD or HeFH. The onset of the pharmacodynamic effects of inclisiran after subcutaneous administration is rapid, with sustained and durable effects (>6 months) observed. Due to the unique characteristic of a disconnection between inclisiran PK and PD responses, the population modeling efforts focused on the PD of inclisiran rather than PK.

Population K-PD modeling was used to characterize inclisiran's dose-related therapeutic effects, and to evaluate the impact of intrinsic and extrinsic factors on clinical response. The population K-PD model was developed based on the inclisiran mechanism of action. A hypothetical liver effect site compartment was linked to the zero-order synthesis rate of PCSK9 through an I_{max} model. PCSK9 levels were then linked to the first-order degradation/clearance rate of LDL-C through the use of a second I_{max} function. The typical elimination half-life in the effect compartment was estimated to be approximately 94 days for subjects with ASCVD, and 78 days for subjects with HeFH, supporting the long duration of effect and infrequent dosing schedule of inclisiran. A dose-dependent reduction in PCSK9 and LDL-C was observed following doses of up to 300 mg. Doses greater than 300 mg did not provide additional effects on reducing LDL-C levels, indicating that the maximal effect of inclisiran was reached at 300 mg. Therefore, the dose regimens of inclisiran impacting the magnitude of LDL-C lowering may be adjusted by changing inclisiran dose frequency, rather than increasing doses to greater than 300 mg.

A covariate analysis explored 120 covariate-PD relationships, including body weight, age, gender, statin coadministration, and population (ie, ASCVD versus HeFH versus subjects with renal or hepatic impairment versus healthy subjects) on inclisiran PD parameters. Some

covariates demonstrated to have effects on IC50_P and on IC50_L, such as observed baseline levels of PCSK9 and LDL-C, and population. This resulted in differences on treatment PD levels in the two populations (ASCVD versus HeFH) with lower percent change reductions from baseline in subjects with HeFH at a 300 mg dose every 6 months. Other covariate effects were deemed not clinically relevant; therefore, no dose adjustments are recommended in any population based on demographics or concomitant statin use.

Population K-PD model simulations were conducted to evaluate potential inclisiran dosing regimens. These simulations demonstrated that maximum LDL-C reductions are observed at approximately 27 days after dosing, and steady-state LDL-C lowering was more effectively achieved with a second dose administered 3 months after the first dose, then followed by every 6 months dosing. This dosing regimen results in rapid and sustained LDL-C lowering and minimal fluctuation of LDL-C levels over time, and this was the dosing regimen that was used in Phase 3 pivotal clinical trials and approved for clinical practice in 2020 by the EMA and in 2021 by the FDA.

Although baseline LDL-C levels differed between subjects with ASCVD and HeFH, simulated absolute changes from baseline were comparable, with absolute LDL-C reductions of approximately 49 mg/dL in both populations, based on the medians of the annual mean. After correction for the higher baseline levels of LDL-C in subjects with HeFH, the percentage LDL-C changes from baseline is expected to be lower in this population in comparison to subjects with ASCVD with a median steady-state pre-dose level of -42.4% in ASCVD versus -27.8% in HeFH populations, a median nadir of -56.7% ASCVD versus -46.1% in HeFH populations, and a median annual mean level of -49.7% in ASCVD versus -38.2% in HeFH populations.

Population simulations demonstrated that the Phase III dose regimen (i.e., 300 mg inclisiran sodium on Day 1, on Day 90, and every 6 months) delivers a rapid and sustained reduction in LDL-C levels during treatment, with median levels of LDL-C below 70 mg/dL in subjects with ASCVD, and below 100 mg/dL in subjects with HeFH. Based on the population K-PD model, the dose frequency of every 6 months results in minimal fluctuations in pre-dose and nadir LDL-C levels at steady state, ensuring sustained and stable efficacy for patients receiving the treatment of inclisiran.

In summary, the population K-PD model established for inclisiran robustly characterized the dose-response and time-course of inclisiran mediated effects on PCSK9 and LDL-C lowering leveraging clinical data from more than 4000 subjects. The use of modeling and simulation was successful in facilitating clinical development of inclisiran, by guiding the study design and by selecting one dosing regimen for Phase III clinical trials initially based on limited clinical data from Phase I and Phase II trials. The population K-PD model was constructed based on the mechanism of action of inclisiran, and focused on PD only, due to the disconnection between systemic PK exposure and PD, which is a unique aspect of the GalNac-conjugated siRNA therapy. The established population K-PD model was also used to address a variety of questions from the health authorities such as EMA and FDA attributing to its novel nature of first-in-class siRNA therapy for a chronic disease, playing a key role in obtaining the regulatory approval of inclisiran. In addition, this population K-PD model was used to inform the inclisiran label for managing dosing instructions for the health provider and patients in situation where patients may experience missed or delayed doses of inclisiran, which may be proven to be especially helpful during the current pandemic period of COVID-19. The overall timeline for clinical development of this first novel siRNA therapy for the treatment of a chronic disease condition, from the first-

in-human clinical study to the first NDA approval, was six years (2014-2020), and we believe that the timely and sensible use of population K-PD modeling and simulation approach played a key role in successfully selecting the one dosing regimen for Phase III clinical trials based on limited Phase I and II clinical data.

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Author Contributions

- Nathalie H Gosselin, Virna J.A Schuck, Olivier Barriere, Kenneth Kulmatycki, Alison Margolskee, Patrick Smith and YanLing He wrote the manuscript;
- YanLing He designed the research;
- Nathalie H Gosselin, Virna J.A Schuck, Olivier Barriere and Alison Margolskee performed the research;
- Nathalie H Gosselin, Virna J.A Schuck, Olivier Barriere, Kenneth Kulmatycki, Alison Margolskee, Patrick Smith and YanLing He analyzed the data.

Study Highlights

o What is the current knowledge on the topic?

Inclisiran is siRNA therapy designed to specifically target PCSK9 mRNA in the liver for the treatment of hypercholesterolemia by reducing LDL-C, a causal and modifiable factor in the pathogenesis of ASCVD.

o What question did this study address?

Establish the population K-PD model of inclisiran and demonstrate the utility of the K-PD model in determining the dosing regimen for Phase 3 clinical development of inclisiran.

o What does this study add to our knowledge?

This study of translational population K-PD modeling approach provides evidence that the dosing regimen of an siRNA therapy could be confidently and reliably identified based on K-PD modeling. Due to disconnection between PK and PD of the liver-targeted siRNA therapy, modeling approach of PD endpoint only is proved to be useful.

o How might this change clinical pharmacology or translational science?

The timely and sensible use of modeling and simulation approach from data obtained during Phase 1 and 2 to the successful Phase 3, then become a medicine to address unmet medical needs for patients, is the key aspect of translational science.

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Table 1 Summary of Covariates at Baseline by Patient Population. Continuous Covariates Presented as Mean (SD). Categorical Covariates Presented as N (% Total)

Continuous Covariates						
	Phase 1 Population (N=129)	Renal Impairment (N=31)	Hepatic Impairment (N=16)	ASCVD or ASCVD-risk equivalent (N=3671)	HeFH (N=481)	Overall (N=4328)
WT (kg)	77.5 (12.4)	82.5 (14.4)	87.8 (13.2)	89.2 (19.2) ^a	83.9 (18.4)	88.2 (19.1) ^b
Age (years)	45.4 (13.7)	56.3 (15.2)	54.2 (12.4)	65.1 (9.08)	54.7 (12.1)	63.3 (10.7)
eGFR (mL/min/1.73 m ²)	90.7 (17.8)	51.5 (33.0)	103 (20.7)	78.1 (20.7)	85.1 (19.9)	79.2 (21.0)
Bilirubin (mg/dL)	0.513 (0.216)	0.504 (0.269)	2.28 (2.16)	0.624 (0.265)	0.671 (0.339)	0.631 (0.318)
PCSK9 (ng/mL)	287 (85.3)	286 (84.0)	279 (124)	392 (141)	441 (134)	393 (141)
LDL-C (mg/dL)	124 (37.5)	121 (43.2)	121 (109)	108 (41.3)	153 (54.3)	114 (45.5)
Triglycerides (mg/dL)	113 (64.6)	128 (78.7)	111 (63.7)	151 (81.0)	135 (69.3)	147 (79.7)
Categorical Covariates						
	Phase 1 Population (N=129)	Renal Impairment (N=31)	Hepatic Impairment (N=16)	ASCVD or ASCVD- risk equivalent (N=3671)	HeFH (N=481)	Overall (N=4328)
Gender						
Male	87 (67.4%)	22 (71.0%)	10 (62.5%)	2564 (69.8%)	227 (47.2%)	2910 (67.2%)
Female	42 (32.6%)	9 (29.0%)	6 (37.5%)	1107 (30.2%)	254 (52.8%)	1418 (32.8%)
Race						
White	85 (65.9%)	22 (71.0%)	15 (93.8%)	3382 (92.1%)	452 (94.0%)	3956 (91.4%)
Black	33 (25.6%)	0 (0%)	0 (0%)	235 (6.4%)	15 (3.1%)	283 (6.5%)
Asian	6 (4.7%)	3 (9.7%)	0 (0%)	27 (0.7%)	12 (2.5%)	48 (1.1%)
Others	5 (3.9%)	1 (3.2%)	0 (0%)	2 (0.1%)	0 (0%)	8 (0.2%)
Native Hawaiian or other Pacific Islander	0 (0%)	5 (16.1%)	1 (6.2%)	12 (0.3%)	1 (0.2%)	19 (0.4%)
American Indian or Alaska Native	0 (0%)	0 (0%)	0 (0%)	13 (0.4%)	1 (0.2%)	14 (0.3%)
Diabetic Status						
Nondiabetic	129 (100%)	27 (87.1%)	13 (81.2%)	2665 (72.6%)	438 (91.1%)	3272 (75.6%)
Diabetic	0 (0%)	4 (12.9%)	3 (18.8%)	1004 (27.3%) ^c	41 (8.5%) ^d	1052 (24.3%) ^e
Statin Category						
Absence	116 (89.9%)	20 (64.5%)	13 (81.2%)	680 (18.5%)	121 (25.2%)	950 (22.0%)

Table 1 Summary of Covariates at Baseline by Patient Population. Continuous Covariates Presented as Mean (SD). Categorical Covariates Presented as N (% Total)

	Continuous Covariates					
	Phase 1 Population (N=129)	Renal Impairment (N=31)	Hepatic Impairment (N=16)	ASCVD or ASCVD-risk equivalent (N=3671)	HeFH (N=481)	Overall (N=4328)
Presence	13 (10.1%)	11 (35.5%)	3 (18.8%)	2991 (81.5%)	360 (74.8%)	3378 (78.0%)

a: N=3669; b: N=4326; c: N=3669; d: N=479; e: N=4324

ASCVD=atherosclerotic cardiovascular disease; ALT=alanine aminotransferase; AST=aspartate aminotransferase; BMI=body mass index; CRCL=creatinine clearance; eGFR=estimated glomerular filtration rate; HeFH=heterozygous familial hypercholesterolemia; LDL-C=low-density lipoprotein cholesterol; N=number of subjects; PCSK9=proprotein convertase subtilisin/kexin type 9; SD=standard deviation; WT=body weight

Note: Classification of population was based on trials (i.e., Phase 1 population [ALNPCSSC-001, healthy subjects in ORION-6 and ORION-12] versus renal impairment [ORION-7] versus hepatic impairment [ORION-6] versus ASCVD or ASCVD-risk equivalent [ORION-1, ORION-3, ORION-10 and ORION-11] versus HeFH [ORION-9]).

Table 2: Final Population K-PD Model of PCSK9 and LDL-C

Parameters	Typical Value of Final Model	%RSE	%IIV (%Shrinkage)
Ke (/day) (subjects with ASCVD)	0.007356	1.26	43.9 (29.3)
× (1+ θ) for HeFH	0.2077	28.0	
× (1+ θ) for Others	-0.2089	24.4	
× (WT/86) ⁰	0.3375	14.8	
Baseline PCSK9 (ng/mL)	384.3	0.203	10.8 (19.3)
× (obsPCSK9/376) ⁰	0.7088	1.70	
Kdeg_P (/day)	0.1665	2.38	
IC50_P (mg) (subjects with ASCVD and Others)	44.37	2.88	92.9 (13.5)
× (BIL/0.57) ⁰	-0.4228	5.25	
× (1+ θ) for HeFH	1.172	12.9	
× (obsPCSK9/376) ⁰	-1.315	3.58	
× (1+ θ) for absence of statin	-0.1884	20.4	
× (WT/86) ⁰	-0.5216	17.6	
Imax_P	0.8817	0.287	
Baseline LDL-C (mg/dL)	104.4	0.196	13.2 (12.4)
× (age [years]/65) ⁰	-0.1706	16.6	
× (obsLDL-C/103) ⁰	0.7592	1.77	
Ksyn_L (mg/dL/day)	16.78	6.58	
IC50_L (ng/mL) (subjects with ASCVD)	112.4	3.66	147 (17.4)
× (age [years]/65) ⁰	-0.8275	16.3	
× (1+ θ) for diabetic	-0.2464	18.4	
× (1+ θ) for HeFH	0.1524	56.1	
× (1+ θ) for Others	-0.1647	51.7	
× (obsLDL-C/103) ⁰	-0.4651	35.8	
× (1+ θ) for female	0.2069	29.7	
× (1+ θ) for absence of statin	0.5264	20.1	
× (TRIG/128) ⁰	-0.6153	18.5	
× (WT/86) ⁰	-0.5572	26.1	
Imax_L	1 fixed	0	
Residual error			
Proportional error PCSK9	0.1966	0.776	

Parameters	Typical Value of Final Model	%RSE	%IIV (%Shrinkage)
Additive error LDL-C (mg/dL)	6.708	4.44	
Proportional error LDL-C	0.1822	1.58	

Note 1: placebo subjects were not considered in the shrinkage calculation for PD parameters related to inclisiran.

Note 2: HeFH subjects were from ORION-9 only.

Note 3: Others correspond to subjects in Phase 1 studies, subjects with renal impairment or subjects with hepatic impairment.

Note 4: Final model was based on dataset with exclusion criteria #2.

BIL=baseline total bilirubin (mg/dL); HeFH=heterozygous familial hypercholesterolemia; IC_{50L}=concentration of PCSK9 to achieve 50% of maximum inhibitory effect on LDL-C; IC_{50P}=concentration in the effect compartment to achieve 50% of maximum inhibitory effect on PCSK9; I_{maxL}=maximum inhibitory effect of inclisiran on LDL-C levels; I_{maxP}=maximum inhibitory effect of inclisiran on PCSK9 levels; K_{degP}=degradation rate of PCSK9; K_e=first-order rate constant for effect compartment; K_{synL}=synthesis rate of LDL-C; LDL-C=low-density lipoprotein cholesterol; obsLDL-C= baseline of observed LDL-C (mg/dL); obsPCSK9= baseline of observed PCSK9 (ng/mL); PCSK9=proprotein convertase subtilisin/kexin type 9; PD=pharmacodynamic; RSE=relative standard error; TRIG=baseline triglycerides (mg/dL); WT= baseline body weight (kg).

Table 3: PCSK9 and LDL-C Response at Steady State in a Virtual Population Receiving 300 mg SC Inclisiran Sodium Injection on Day 1, on Day 90, and every 6 months for 4 Years

	ASCVD or ASCVD-risk equivalent		HeFH	
	PCSK9 (ng/mL) (ng×day/mL)	LDL-C (mg/dL) (mg×day/dL)	PCSK9 (ng/mL) (ng×day/mL)	LDL-C (mg/dL) (mg×day/dL)
Steady-State Predose Levels				
Median [90% PI]	147 [60.0, 328]	58.0 [25.9, 120]	223 [94.7, 417]	93.0 [46.0, 169]
%Change from Baseline – Steady-State Predose Levels				
Median [90% PI]	-61.1 [-82.3, -18.4]	-42.4 [-69.8, -10.5]	-45.6 [-77.3, -12.7]	-27.8 [-63.0, -6.16]
Nadir				
Median [90% PI]	84.9 [47.2, 177]	45.4 [18.8, 104]	122 [67.2, 255]	72.6 [32.6, 152]
%Change from Baseline – Nadir				
Median [90% PI]	-77.7 [-85.7, -54.8]	-56.7 [-77.0, -21.6]	-70.8 [-83.8, -44.9]	-46.1 [-73.0, -15.2]
Fluctuations				
Median [90% PI]	56.3 [9.97, 153]	9.98 [1.64, 36.4]	85.4 [21.3, 203]	16.6 [3.38, 46.4]
%Change from baseline – fluctuations				
Median [90% PI]	14.8 [2.94, 39.8]	9.41 [1.62, 31.1]	20.6 [5.67, 43.8]	11.9 [2.63, 33.1]
Annual Mean				
Median [90% PI]	198 [35.9, 532]	34.6 [5.66, 117]	290 [74.9, 681]	55.2 [12.2, 158]
%Change from Baseline – Annual Mean				
Median [90% PI]	-70.8 [-84.2, -39.6]	-49.7 [-73.9, -19.4]	-59.9 [-80.8, -29.9]	-38.2 [-67.3, -11.4]

The PD parameters were derived based on the time profiles of PCSK9 and LDL-C after the dose at Day 450.

Nadir is the lowest value; trough is the value immediately prior to next dose; annual mean is average concentration during the last year of treatment.

AUEC=area under the effect concentration; ASCVD=atherosclerotic cardiovascular disease; CV=coefficient of variation; HeFH=heterozygous familial hypercholesterolemia; LDL-C=low-density lipoprotein cholesterol;

PCSK9=proprotein convertase subtilisin/kexin type 9; PD=pharmacodynamic; PI=prediction interval;

SC=subcutaneous.

Note: Classification of population was based on trials (i.e., ASCVD or ASCVD-risk equivalent [ORION-1, ORION-3, ORION-10 and ORION-11] versus HeFH [ORION-9]).

FIGURES

Figure 1 Schematic Representation of the Base Structural Model for PCSK9 and LDL C after SC Injection of Inclisiran

The full lines represent the median simulated levels and shaded area the 95%PI of simulated levels.

K_e =first-order rate constant for effect compartment; K_{degL} =degradation rate of LDL-C; K_{degP} =degradation rate of PCSK9; K_{synL} =synthesis rate of LDL-C; K_{synP} =synthesis rate of PCSK9; LDL-C=low-density lipoprotein cholesterol; PCSK9=proprotein convertase subtilisin/kexin type 9; PI=prediction interval

Figure 2 Predicted Levels [Median $\pm$ 90% PI] Compared with Observed Levels [Median $\pm$ 90 PI]

ASCVD=atherosclerotic cardiovascular disease; HeFH=heterozygous familial hypercholesterolemia; LDL-C=low-density lipoprotein cholesterol; PCSK9=proprotein convertase subtilisin/kexin type 9; PI=prediction interval.

Note: Classification of population was based on trials (i.e., ASCVD or ASCVD-risk equivalent [ORION-1, ORION-3, ORION-10 and ORION-11] versus HeFH [ORION-9]).

Figure 3: Covariate Effects on Annual Mean Levels of LDL-C (mg/dL) in Subjects with ASCVD or ASCVD-Risk Equivalent and Subjects with HeFH

The dots and the horizontal blue segments represent the bootstrap-derived median and 95% CI of annual mean levels of LDL-C based on covariates observed in subjects with ASCVD or ASCVD-risk equivalent or with HeFH (2.5th, 50th, and 97.5th percentiles) and dashed vertical lines the values for the reference subjects within each sub population.

ASCVD=atherosclerotic cardiovascular disease; CI=confidence interval; HeFH=heterozygous familial hypercholesterolemia; LDL-C=low-density lipoprotein cholesterol; PCSK9=proprotein convertase subtilisin/kexin type 9.

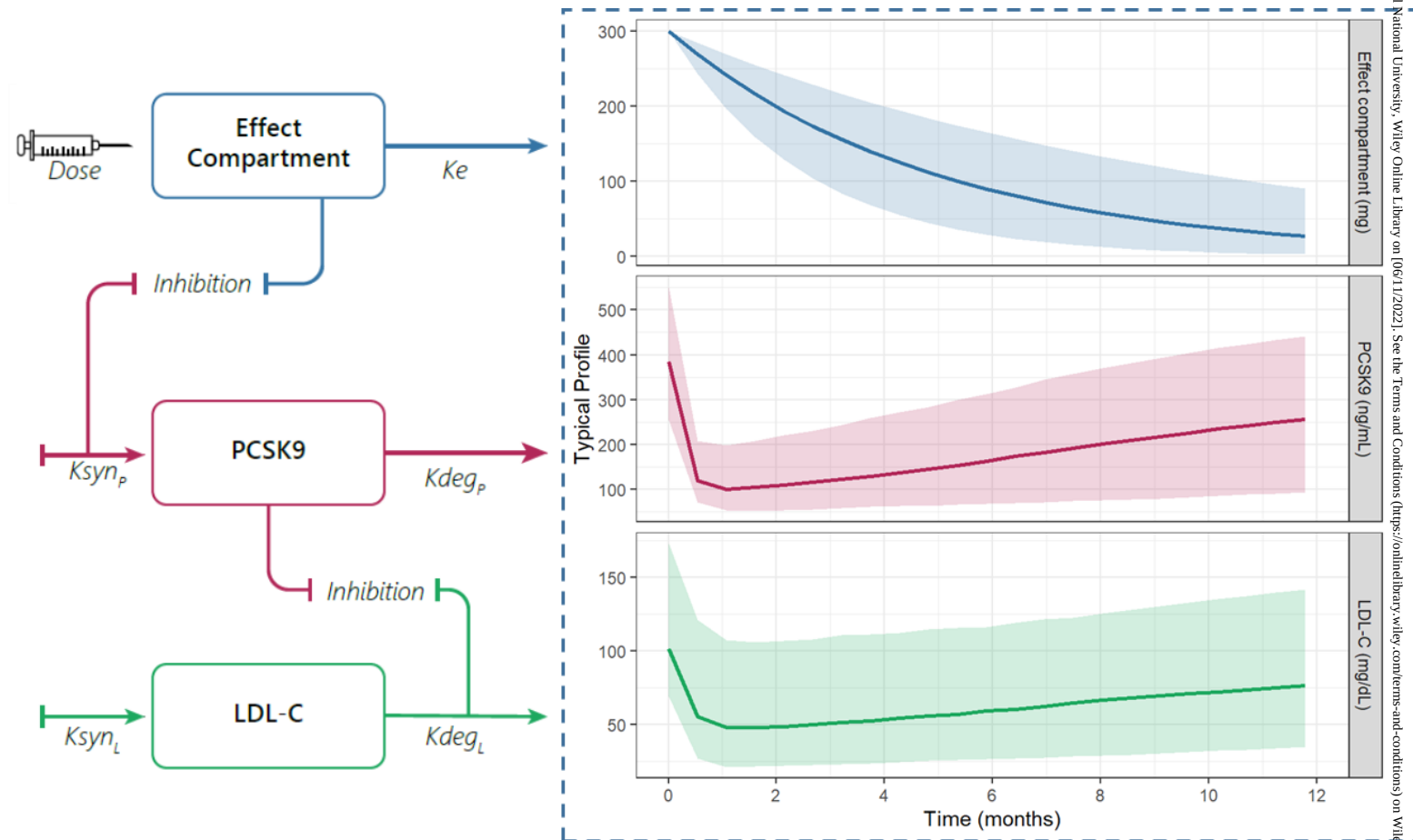
Note 1: Classification of population was based on trials (i.e., ASCVD or ASCVD-risk equivalent [ORION-1, ORION-3, ORION-10 and ORION-11] versus HeFH [ORION-9]).

Note 2: The reference subject with ASCVD or ASCVD-risk equivalent is a non-diabetic male of 66 years-old with bilirubin of 0.59 mg/dL, observed baseline LDL-C of 96 mg/dL, observed baseline PCSK9 of 372 ng/mL, triglycerides of 132 mg/dL, body weight of 88 kg and under statin. The reference subject with HeFH is a non-diabetic female of 56 years-old with bilirubin of 0.62 mg/dL, observed baseline LDL-C of 139 mg/dL, observed baseline PCSK9 of 433 ng/mL, triglycerides of 119 mg/dL, body weight of 81 kg and under statin.

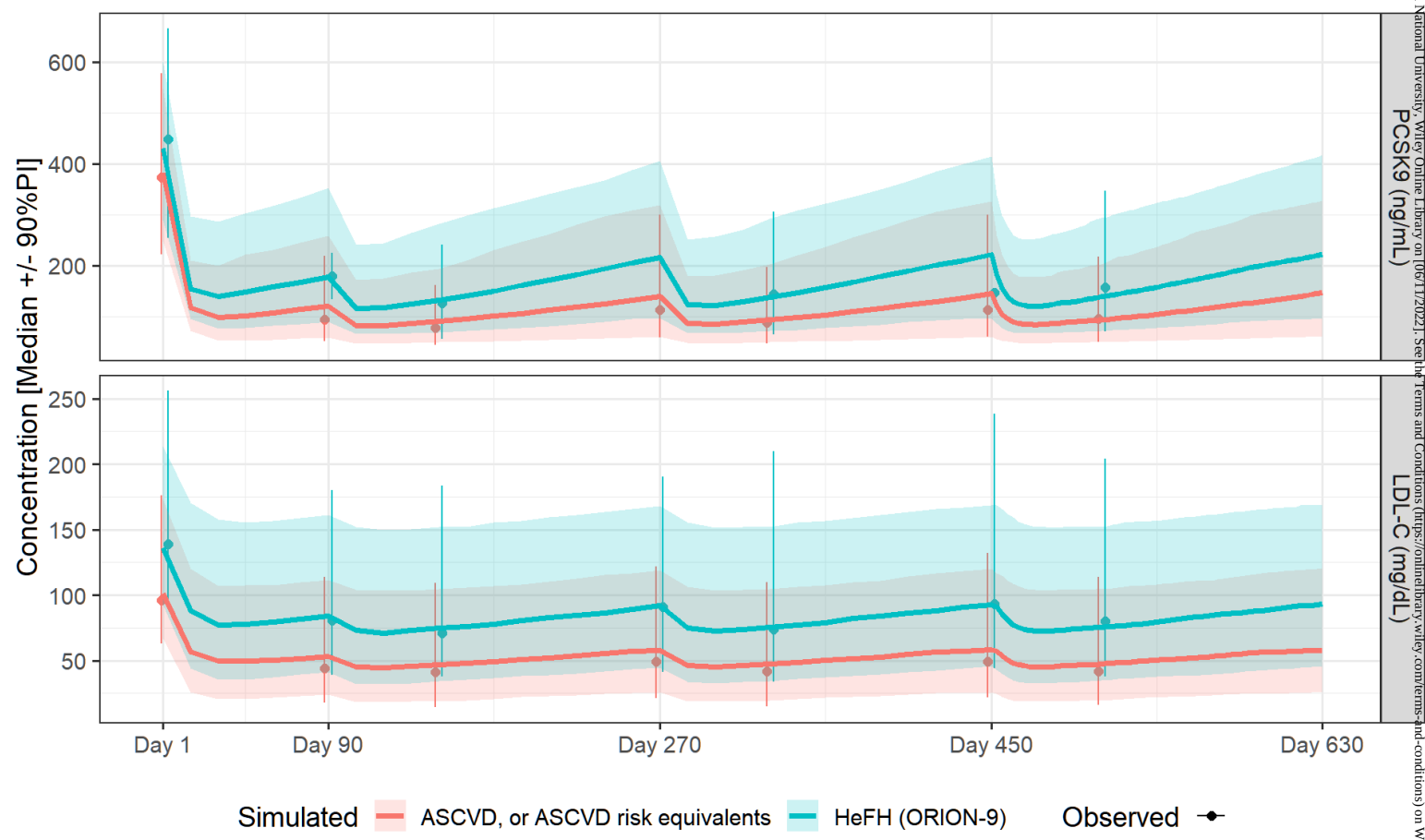
Figure 4 Median PCSK9 and LDL-C Levels in Virtual Subjects with ASCVD or ASCVD-Risk Equivalent Receiving 300 mg SC Inclisiran Sodium Injection For Two Years With a Dose Delay or Missing Dose on Month 21 of Treatment

ASCVD=atherosclerotic cardiovascular disease; LDL-C=low-density lipoprotein cholesterol; PCSK9=proprotein convertase subtilisin/kexin type 9; SC=subcutaneous.

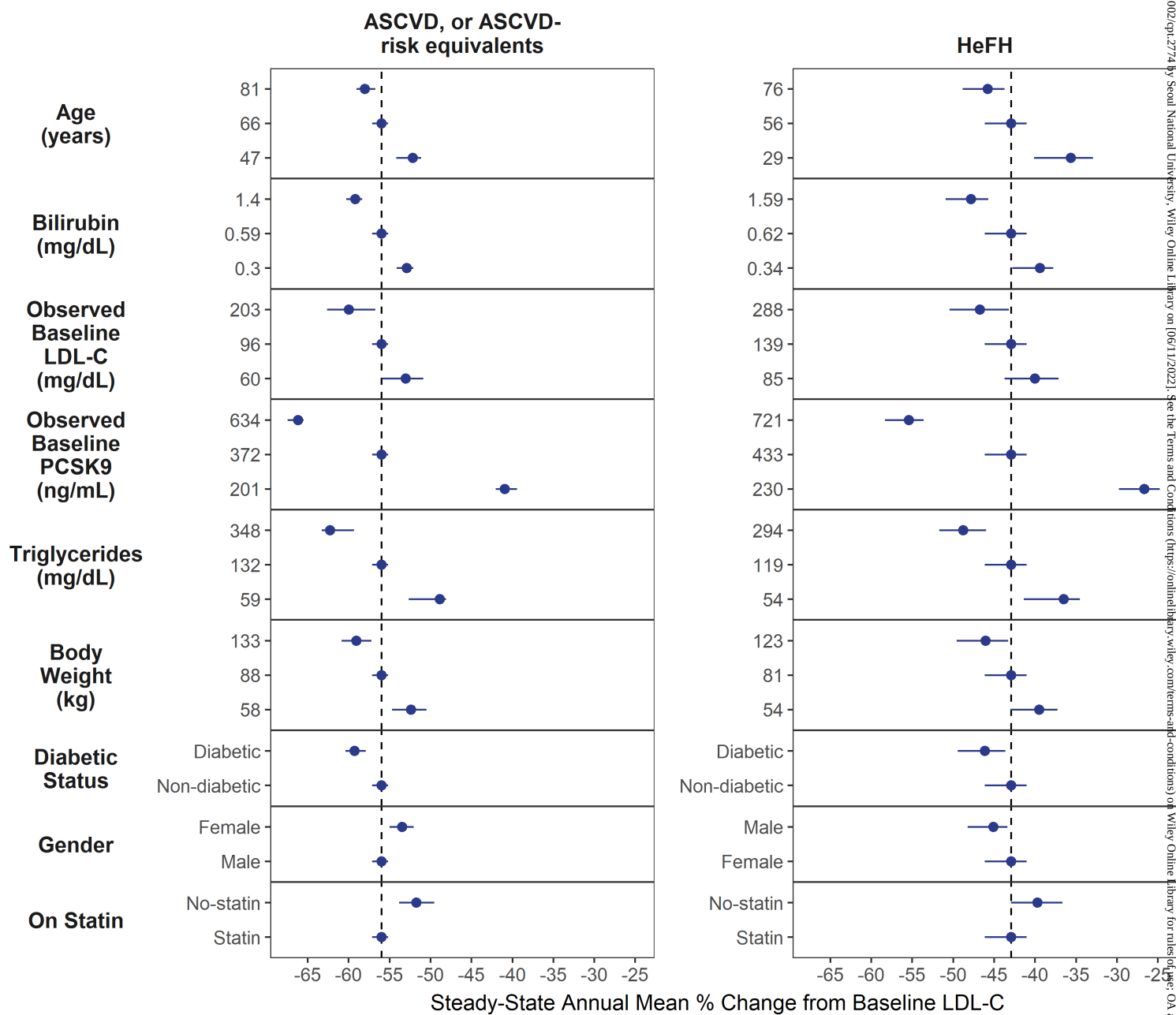
Note: Classification of population was based on trials (i.e., ASCVD or ASCVD-risk equivalent [ORION-1, ORION-3, ORION-10 and ORION-11]).



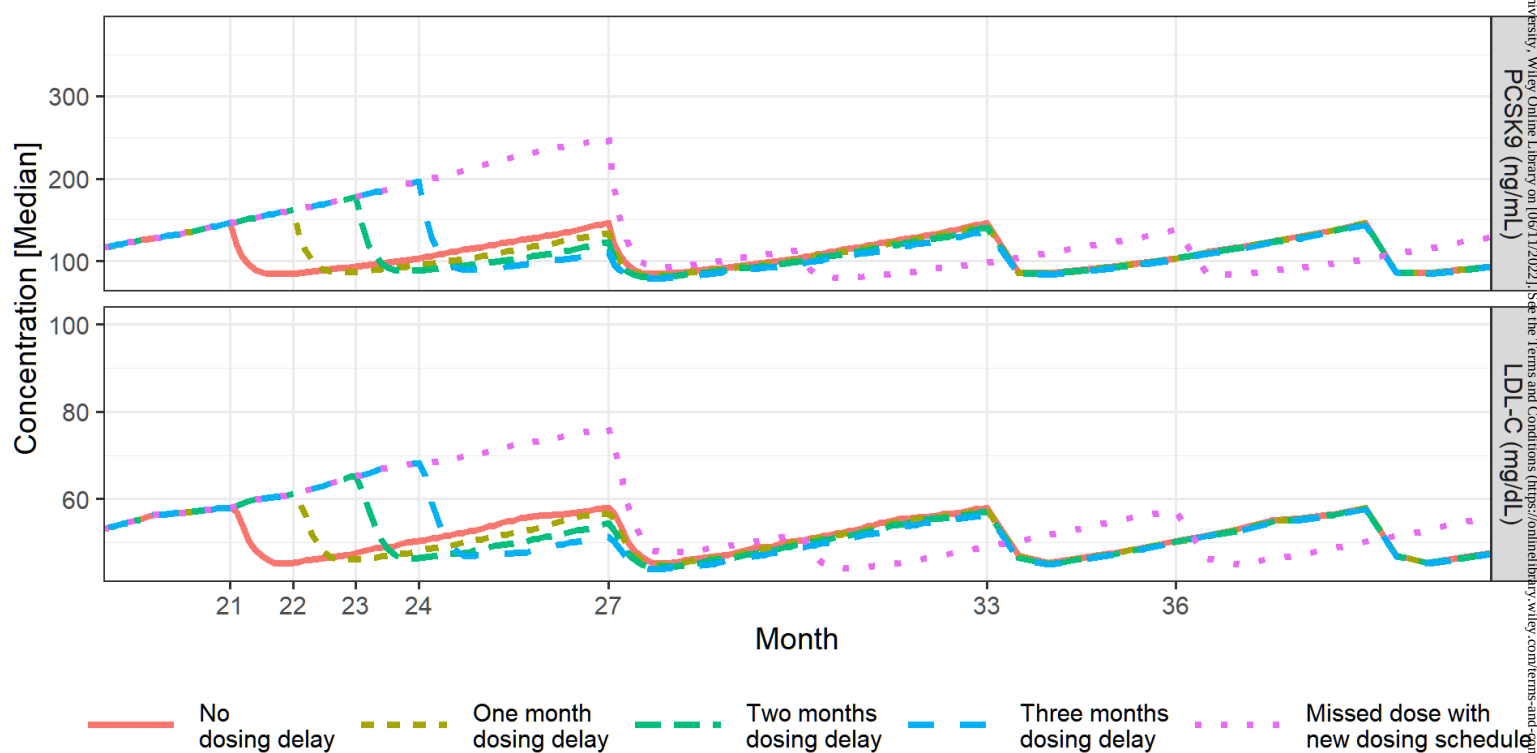
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